MATHEMATICS

SOLUTION OF TRIANGLE

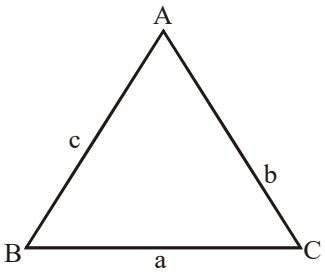
THEORY SHEET

ENGLISH

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1. INTRODUCTION ABOUT TRIANGLE

A triangle has three sides and three angles.



(i) $A + B + C = 180^\circ$ or π (ii) $a + b > c,\ b + c > a,\ c + a > b$

(iii) $a > 0,\ b > 0,\ c > 0$ (iv) $|a-b| < c,\ |a-c| < b,\ |b-c| < a$

(v) If $A = B \Leftrightarrow a = b$ (vi) If $A > B \Leftrightarrow a > b$

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

2. BASIC LAWS

2.1 SINE LAW

The sides of a triangle (any type of triangle) are proportional to the sines of the angle opposite to them.

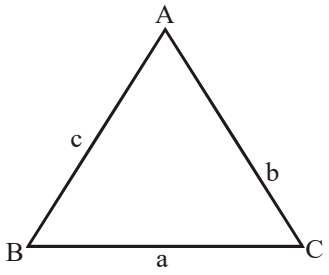
In triangle ABC $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C} = k = 2R$

2.2 COSINE RULE

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc};$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca};$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



Proof :

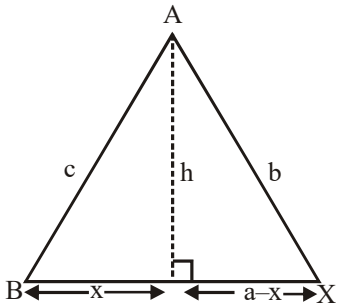
$h^2 = c^2 - x^2$ $\qquad h^2 = b^2 - (a-x)^2$

$c^2 - x^2 = b^2 - (a-x)^2$

$c^2 - x^2 = b^2 - (a^2 + x^2 - 2ax)$ $\qquad (x = c\cos B)$

$c^2 + a^2 - b^2 = 2ac\cos B$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}$$



Note

1. cosine rule is to be used when
   (i) All three sides are given.
   (ii) Two sides and angle between them is given.
2. We can determine nature of a triangle when all 3 sides are known
   Let a,b,c are sides and c is largest side; then C is largest angle ;

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

If $a^2 + b^2 > c^2 \Rightarrow \Delta$ is acute angled triangle

$a^2 + b^2 = c^2 \Rightarrow \Delta$ is right angled triangle

$a^2 + b^2 < c^2 \Rightarrow \Delta$ is obtuse angled triangle.

2.3 PROJECTION FORMULAE

In any ΔABC ;

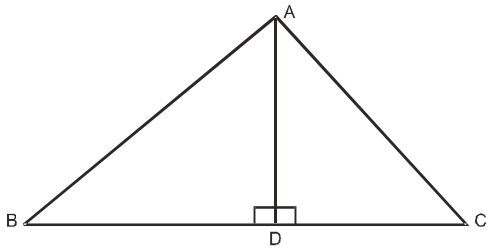
(i) $a = b \cos C + c \cos B$ (ii) $b = c \cos A + a \cos C$ (iii) $c = a \cos B + b \cos A$

i.e. any side of a triangle is equal to the sum of the projection of other two sides on it.

Proof:

$BC = BD + CD$

$a = b \cos C + c \cos B$



2.4 BASIC LAWS : TANGENT RULE

(a) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$ **(b)** $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a}\cot\frac{B}{2}$ **(c)** $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

Note

Tangent rule is to applied when two sides and angle between then is known.

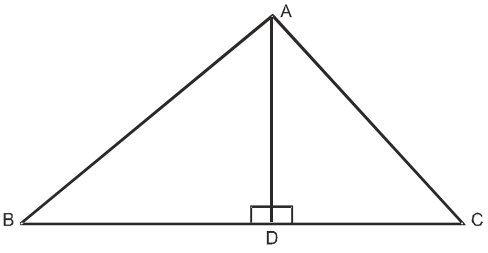
2.5 AREA OF TRIANGLE

If Δ represents the area of a triangle ABC, then area of

$$\Delta = \frac{1}{2} BC \cdot AD, \quad \left[\because \Delta = \frac{1}{2}(\text{base})(\text{height})\right]$$

$$= \frac{1}{2} a \cdot (c \sin B) \quad \left\{\text{as } \sin B = \frac{AD}{c}\right\}$$

$$\Delta = \frac{1}{2} ac \cdot \sin B$$



$$\Delta = \frac{1}{2} ab \sin C = \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B$$

Questions

1. Prove that $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) = 0$

Sol. P.T. $a \sin(B - C) + b \sin(C - A) + C \sin(A - B) = 0$

$\Sigma K \sin A \sin(B - C)$

$= \Sigma K(\sin A \sin B \cos C - \sin A \sin C \cos B)$

$= K \times 0 = 0$

2. Prove that $\sin(B-C) = \dfrac{b^2-c^2}{a^2}\sin A$

Sol. RHS $\Rightarrow \dfrac{4R^2\sin^2 B - 4R^2\sin^2 C}{4R^2\sin^2 A} \times \sin A$

$\Rightarrow 4R^2 \dfrac{\sin(B+C)\sin(B-C)}{4R^2\sin^2 A} \times \sin A = \sin(B-C)$

3. Prove that $a\cos\left(\dfrac{B-C}{2}\right) = (b+c)\sin\dfrac{A}{2}$

Sol. RHS $k\times(\sin B+\sin C)\sin\left(\dfrac{A}{2}\right)$

$k\times 2\sin\left(\dfrac{B+C}{2}\right)\cos\left(\dfrac{B-C}{2}\right)\cos\left(\dfrac{B+C}{2}\right)$

$k\sin(B+C)\cos\left(\dfrac{B-C}{2}\right)$

$k\sin A\cos\left(\dfrac{B-C}{2}\right) = a\cos\left(\dfrac{B-C}{2}\right)$

4. Prove that $(b^2-c^2)\cot A + (c^2-a^2)\cot B + (a^2-b^2)\cot C = 0$

Sol. Since $a = k\sin A$, $b = k\sin B$ and $c = k\sin C$

$\therefore$ $(b^2 - c^2)\cot A = k^2(\sin^2 B - \sin^2 C)\cot A = k^2\sin(B+C)\sin(B-C)\cot A$

$\therefore$ $= k^2\sin A\sin(B-C)\dfrac{\cos A}{\sin A}$

$= -k^2\sin(B-C)\cos(B+C)$ ($\because \cos A = -\cos(B+C)$)

$= -\dfrac{k^2}{2}[2\sin(B-C)\cos(B+C)]$

$= -\dfrac{k^2}{2}[\sin 2B - \sin 2C]$ .....(i)

Similarly $(c^2 - a^2)\cot B = -\dfrac{k^2}{2}[\sin 2C - \sin 2A]$ .....(ii)

and $(a^2 - b^2)\cot C = -\dfrac{k^2}{2}[\sin 2A - \sin 2B]$ .....(iii)

adding equations (i), (ii) and (iii), we get

$(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C = 0$ **Hence Proved**

5. If a, b, c are in A.P. then prove that sin A, sin B, sin C will be in A.P.

Sol. a, b, c are in AP

$b = \dfrac{a+c}{2}$

$$2R\sin B = \frac{2R\sin A + 2R\sin C}{2} \Rightarrow \sin B = \frac{\sin A + \sin C}{2}$$

6. Prove that $\sin A + \sin B > \sin C$.

Sol. $\sin A + \sin B - \sin C > 0$

LHS $2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) - \sin\frac{C}{2}\cos\frac{C}{2}$

$$2\cos\frac{C}{2}\left(\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right)$$

$$2\cos\frac{C}{2}\times 2\sin\frac{A}{2}\sin\frac{B}{2}$$

+ve × +ve × +ve > 0

7. If $a\cos A = b\cos B$ prove that $A = B$ or $\angle C = 90^\circ$.

Sol. $2R\sin A\cos A = 2R\sin B\cos B$

$\sin 2A = \sin 2B$

$2A = 2B$ or $A + B = \frac{\pi}{2}$

$A = B$ $C = \frac{\pi}{2}$

8. If $A : B : C = 1 : 2 : 3$ find $a : b : c$

Sol. $A = 30^\circ$, $B = 60^\circ$, $C = 90^\circ$

$a = 2R\sin 30^\circ = R$

$b = 2R\sin 60^\circ = \sqrt{3}R$

$c = 2R\sin 90^\circ = 2R$

$a : b : c \Rightarrow 1 : \sqrt{3} : 2$

9. Prove that $\cot A + \cot B + \cot C = \frac{a^2+b^2+c^2}{4\Delta}$.

Sol. $\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}$

$$= \frac{b^2+c^2-a^2}{2bc\sin A} + \frac{a^2+c^2-b^2}{2ac\sin B} + \frac{a^2+b^2-c^2}{2ab\sin C}$$

$$= \frac{b^2+c^2-a^2}{4\Delta} + \frac{a^2+c^2-b^2}{4\Delta} + \frac{a^2+b^2-c^2}{4\Delta} = \frac{a^2+b^2+c^2}{4\Delta}$$

10. If a^2, b^2, c^2, are in AP then prove that cot A, cot B, cot C are in AP.

Sol. cotA, cotB, cotC, in AP

$\frac{\cos A}{\sin A}, \frac{\cos B}{\sin B}, \frac{\cos C}{\sin C}$ in AP

$\frac{b^2+c^2-a^2}{2bc\sin A}, \frac{a^2+c^2-b^2}{2ac\sin B}, \frac{a^2+b^2-c^2}{2ab\sin C}$ in AP

$\frac{b^2+c^2-a^2}{4\Delta}, \frac{a^2+c^2-b^2}{4\Delta}, \frac{a^2+b^2-c^2}{4\Delta}$ in AP

multiply by 4Δ

$b^2 + c^2 - a^2, a^2 + c^2 - b^2, a^2 + b^2 - c^2$ in AP

Subtract $(a^2 + b^2 + c^2)$

$-2a_1^2 - 2b_1^2 - 2c^2$ in AP

divide by (–2)

a^2, b^2, c^2 in AP

11. $b = \sqrt{3}$, $c = 1$, $A = 30^\circ$ solve the triangle.

Sol. $\cos A = \frac{3+1-a^2}{2\times\sqrt{3}\times1}$

$\Rightarrow a^2 = 1 \Rightarrow a = 1$

$\cos B = \frac{1+1-3}{2\times1\times1} = -\frac{1}{2}$

$B = 120°, A = C = 30°$

12. $a = 2, b = 1 + \sqrt{3}$, $C = 60^\circ$ solve the triangle.

Sol. Using tangent rule $\Rightarrow \tan\left(\frac{A-B}{2}\right) = \frac{2-1-\sqrt{3}}{2+1+\sqrt{3}}(\cot 30°)$

$= \frac{1-\sqrt{3}}{\sqrt{3}+1} = \tan 15°$

$A - B = 30°$

$A + B = 120$

$A = 45, B = 75°$

$\frac{a}{\sin A} = \frac{c}{\sin c}$

$\Rightarrow c = \frac{2\times\frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}} \Rightarrow c = \sqrt{6}$

13. If sides of a triangle are 4, 5, 6 find cos A, cos B, cos C, prove that largest angle = 2. smallest angle.

Sol. $\cos A = \frac{25+36-16}{2\times 30} = \frac{3}{4}$

$\cos B = \frac{16+36-25}{2\times 4\times 6} = \frac{9}{16}$

$\cos C = \frac{16+25-36}{2\times 4\times 5} = \frac{1}{8}$

Proof $C = 2A$

$\cos C = \cos 2A$

$\cos 2A = 2\cos^2 A - 1$

$= 2\times\frac{9}{16} - 1$

$= \frac{2}{16}$

$= \frac{1}{8}$

14. Sides of a triangle are a, b, $\sqrt{a^2+b^2+ab}$. Find largest angle.

Sol. c is largest side

c is largest angle

$\cos C = \frac{a^2b^2 - c^2}{2ab}$

$= \frac{a^2+b^2-a^2-b^2-ab}{2ab} = -\frac{1}{2}$

$C = 120°$

15. If $c^4 - 2(a^2 + b^2)\, c^2 + a^4 + b^4 + a^2b^2 = 0$ find $\angle C$.

Sol. $c^4 - 2a^2c^2 - 2a^2b^2 + a^4 + b^4 + a^2b^2 = 0$

$c^4 - 2(a^2 + b^2)\, c^2 + (a^4 + b^4)^2 = a^2b^2$

$(a^2 + b^2 - c^2) = a^2b^2$

$\left(\frac{a^2+b^2-c^2}{ab}\right)^2 = 1$

$(2\cos c)^2 = 1$

$4\cos^2 c = 1$

$\cos^2 c = \frac{1}{4}$

$c = 60°, 120°$

16. If $\frac{2\cos A}{a}+\frac{\cos B}{b}+\frac{2\cos C}{c}=\frac{a}{bc}+\frac{b}{ac}$ prove that $\angle A = 90^\circ$.

Sol. We have $\frac{2\cos A}{a}+\frac{\cos B}{b}+\frac{2\cos C}{c}=\frac{a}{bc}+\frac{b}{ac}$

Multiplying both sides of abc, we get

$\Rightarrow$ $2bc\cos A + ac\cos B + 2ab\cos C = a^2 + b^2$

$\Rightarrow$ $(b^2 + c^2 - a^2) + \frac{(a^2 + c^2 - b^2)}{2} + (a^2 + b^2 - c^2) = a^2 + b^2$

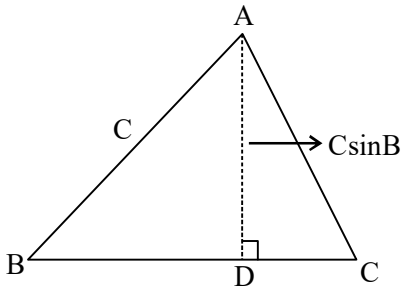
$\Rightarrow$ $c^2 + a^2 - b^2 = 2a^2 - 2b^2$ $\Rightarrow$ $b^2 + c^2 = a^2$

$\therefore$ ΔABC is right angled at A. $\Rightarrow$ $\angle A = 90^\circ$ **Ans. (A)**

17. In ABC AD is altitude and $\angle C = 21^\circ$, $AD = \frac{abc}{b^2 - c^2}$ find $\angle B$.

Sol. $AD = C\sin B$

$$C\sin B = \frac{abc}{b^2 - c^2}$$



$$\sin B = \frac{4R^2 \sin A \sin B}{4R^2\left(\sin^2 B - \sin^2 C\right)}$$

$$1 = \frac{\sin A}{\sin(B+C)\sin(B-C)}$$

$\sin(B - C) = 1 \Rightarrow B - C = 90^\circ$

$\angle B = 111^\circ$

18. If $a\cos^2\frac{C}{2} + c\cos^2\frac{A}{2} = \frac{3b}{2}$, prove that a, b, c are in A.P.

Sol. Here, $\frac{c}{2}(1+\cos A)+\frac{a}{2}(1+\cos C)=\frac{3b}{2}$

$\Rightarrow$ $a + c + (c\cos A + a\cos C) = 3b$

$\Rightarrow$ $a + c + b = 3b$ {using projection formula}

$\Rightarrow$ $a + c = 2b$

which shows a, b, c are in A.P.

19. If a^2, b^2, c^2 in A.P. Prove that $\frac{\sin 3B}{\sin B} = \left(\frac{a^2 - c^2}{2ac}\right)^2$.

Sol. $2b^2 = a^2 + c^2$

$$\text{LHS } \frac{\sin B\left(3 - 4\sin^2 B\right)}{\sin B} = 4\cos^2 B - 1$$

$$= 4 \times \left(\frac{a^2 + c^2 - b^2}{2ac}\right)^2 - 1$$

$$= 4\left(\frac{b^2}{2ac}\right)^2 - 1$$

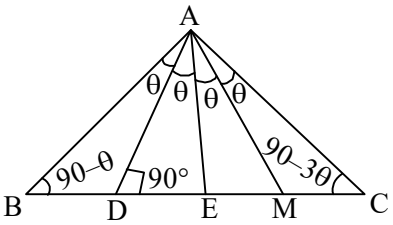
$$= \frac{\left(2b^2\right)^2 - 4a^2c^2}{\left(2ac\right)^2}$$

$$= \frac{\left(a^2 + c^2\right)^2 - 4a^2c^2}{\left(2ac\right)^2}$$

$$= \left(\frac{a^2 - c^2}{2ac}\right)^2$$

20. If altitude, angle bisector and median from vertex A divides $\angle A$ in 4 equal parts find $\angle A$.

Sol.



AD = Altitude

AE = Angle biscatar

AM = Median

ΔABM ΔACM

Sine Rule

$$\frac{AM}{\sin(90 - \theta)} = \frac{BM}{\sin 3\theta} \quad \text{.....(1)} \qquad \frac{AM}{\sin(90 - 3\theta)} = \frac{CM}{\sin\theta} \quad \text{.....(2)}$$

$BM = CM$

eq (1) ÷ (2)

$$\Rightarrow \frac{\cos 3\theta}{\cos\theta} = \frac{\sin\theta}{\sin 3\theta}$$

$\sin 6\theta = \sin 2\theta$

$6\theta + 2\theta = \pi \quad \Rightarrow 8\theta = \pi$

$$\theta = \frac{\pi}{8}$$

$$\angle A = 4\theta = \frac{\pi}{2} = 90^\circ$$

3. HALF ANGLE FORMULAE

$2s = a + b + c$ where s = semi-perimeter of triangle.

$2s - 2a = b + c - a; \quad 2s - 2b = c + a - b; \quad 2s - 2c = a + b - c$

$$\cos A = 2\cos^2\left(\frac{A}{2}\right) - 1 \Rightarrow \qquad 2\cos^2\left(\frac{A}{2}\right) = \cos A + 1$$

$$\Rightarrow 2\cos^2\left(\frac{A}{2}\right) = 1 + \frac{b^2 + c^2 - a^2}{2bc} \qquad = \frac{(b+c)^2 - a^2}{2bc} = \frac{(b+c+a)(b+c-a)}{2bc}$$

$$\Rightarrow 2\cos^2\frac{A}{2} = \frac{(2s)(2s-2a)}{2bc}$$

$$\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}; \qquad \cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}; \qquad \cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$$

$$2\sin\frac{A}{2} = 1 - \cos A$$

$$\Rightarrow \sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}; \quad \sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}; \qquad \sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$$

$$\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} \qquad \tan\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}} \qquad \tan\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$$

$$= \frac{\Delta}{s(s-a)} \qquad = \frac{\Delta}{s(s-b)} \qquad = \frac{\Delta}{s(s-c)}$$

Area of a triangle in terms of sides (Heron's Formula) :

$$\Delta = \frac{1}{2}ab\sin C = \frac{1}{2}ab\left(2\sin\frac{C}{2}\cos\frac{C}{2}\right) = ab\sqrt{\frac{(s-a)(s-b)}{ab}}\sqrt{\frac{s(s-c)}{ab}}$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

Questions

1. Prove that $(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right)=2c\cot\frac{C}{2}$.

Sol. $\because$ L.H.S. $=(a+b+c)\left(\tan\frac{A}{2}+\tan\frac{B}{2}\right)$

$\because$ $\tan\frac{A}{2}=\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ and $\tan\frac{B}{2}=\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

$\therefore$ L.H.S. $=(a+b+c)\left[\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}+\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}\right]$

$=2s\sqrt{\frac{s-c}{s}}\left[\sqrt{\frac{s-b}{s-a}}+\sqrt{\frac{s-a}{s-b}}\right]$

$=2\sqrt{s(s-c)}\left[\frac{s-b+s-a}{\sqrt{(s-a)(s-b)}}\right]$ $\because$ $2s=a+b+c$

$\therefore$ $2s-b-a=c$

$=2\sqrt{s(s-c)}\left[\frac{c}{\sqrt{(s-a)(s-b)}}\right]$ $=2c\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$

$\because$ $\cot\frac{C}{2}=\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$ $=2c\cot\frac{C}{2}=$ R.H.S.

Hence L.H.S. = R.H.S. **Proved**

2. Prove that $(b+c-a)\left(\cot\frac{B}{2}+\cot\frac{C}{2}\right)=2a\cot\frac{A}{2}$.

Sol. LHS $(2s-2a)\left(\frac{s(s-b)+s(s-c)}{\Delta}\right)$

$2s(s-a)\left(\frac{(s-b)+(s-c)}{\Delta}\right)$

$=\frac{2s(s-a)}{\Delta}\times a=2a\cot\frac{A}{2}$

3. Prove that $\frac{\cot\left(\frac{A}{2}\right)+\cot\left(\frac{B}{2}\right)+\cot\left(\frac{C}{2}\right)}{\cot A+\cot B+\cot C}=\frac{(a+b+c)^2}{a^2+b^2+c^2}$.

Sol. $\cos A+\cot B+\cot C$

$=\frac{\cos A}{\sin A}+\frac{\cos B}{\sin B}+\frac{\cos C}{\sin C}$

$$= \frac{b^2+c^2-a^2}{2bc\sin A} + \frac{a^2+c^2-b^2}{2ac\sin B} + \frac{a^2+b^2-c^2}{2ab\sin C}$$

$$= \frac{b^2+c^2-a^2}{4\Delta} + \frac{a^2+c^2-b^2}{4\Delta} + \frac{a^2+b^2-c^2}{4\Delta}$$

$$= \frac{a^2+b^2+c^2}{4\Delta}$$

$$\text{LHS } \frac{\frac{s(s-a)}{\Delta} + \frac{s(s-b)}{\Delta} + \frac{s(s-c)}{\Delta}}{\frac{a^2+b^2+c^2}{4\Delta}}$$

$$= \frac{4s(s-a+s-b+s-c)}{a^2+b^2+c^2}$$

$$= \frac{4s\times s}{a^2+b^2+c^2}$$

$$= \frac{(2s)^2}{a^2+b^2+c^2}$$

$$= \frac{(a+b+c)^2}{(a^2+b^2+c^2)}$$

4. If a, b, c are in A.P. then prove that

(i) $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ are in A.P.

(ii) $\cos A \cot\frac{A}{2}, \cos B \cot\frac{B}{2}, \cos C \cot\frac{C}{2}$ are in A.P.

Sol. (i) $\frac{s(s-a)}{\Delta}, \frac{s(s-b)}{\Delta}, \frac{s(s-c)}{\Delta}$

multiply by $\frac{\Delta}{s}$

s – a, s – b, s – c in AP

subtract s

–a, –b, –c in AP

a, b, c in AP

(ii) $\cos A.\cot\frac{A}{2} = \left(1-2\sin^2\frac{A}{2}\right)\cot\frac{A}{2}$

$$= \cot\frac{A}{2} - 2\sin\frac{A}{2}\cos\frac{A}{2}$$

$= \cot\left(\frac{A}{2}\right) - \sin A$

Prove that $\left(\cot\frac{A}{2} - \sin A\right), \left(\cot\frac{B}{2} - \sin B\right), \left(\cot\frac{C}{2} - \sin C\right)$ in AP

$\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ in AP .....(1)

a, b, c in AP

KsinA, K sinB, KsinC in AP

sinA, sinB, sinC in AP .....(2)

from (1) and (2)

$\cot\frac{A}{2} - \sin A, \cot\frac{B}{2} - \sin B, \cot\frac{C}{2} - \sin C$ in AP

4.1 AMBIGUOUS CASE OF SOLUTION OF TRIANGLE

1. If all three sides are known, we will find all three angles with the help of cosine rule.
2. If all three angles are known, we will find ratio of sides with the help of sine rule.
3. If one side & two angles are known, we will find remaining elements with help of sine rule.
4. If two sides and angle between then are know, we will find remaining elements with the help of cosin rule
5. If two sides and one angle opposite to then are known,(b, c, B are known)

By applying cosine rule, we have $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$\Rightarrow \quad a^2 - (2c\cos B)a + (c^2 - b^2) = 0 \Rightarrow a = c\cos B \pm \sqrt{(c\cos B)^2 - (c^2 - b^2)}$

$\Rightarrow \quad a = c\cos B \pm \sqrt{b^2 - (c\sin B)^2}$

$a_1 = c\cos B + \sqrt{b^2 - (c\sin B)^2}$; $a_2 = c\cos B - \sqrt{b^2 - (c\sin B)^2}$

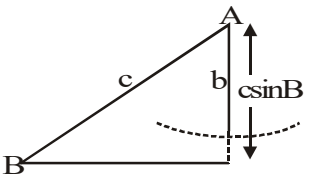
This equation leads to following cases :

1. $b < c\sin B, \Rightarrow D < 0 \Rightarrow$ no such triangle is possible.
2. $b = c\sin B. \Rightarrow D = 0 \Rightarrow$ one positive value of a $\Rightarrow$ one triangle exists
3. $c > b > c\sin B \Rightarrow D > 0 \Rightarrow$ 2 positive values of a $\Rightarrow$ 2 triangles exist
4. $b \geq c \Rightarrow D > 0 \Rightarrow a_1 > 0$ and $a_2 < 0 \Rightarrow$ one triangle exist

Alternate Method

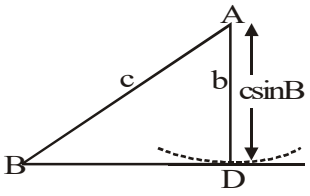
Case I :

$b < c\sin B.$



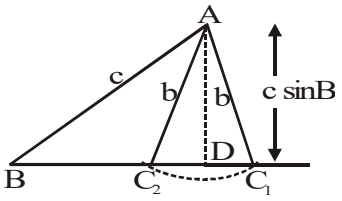
We draw the side c and angle B. Now it is obvious from the figure that there is no triangle possible.

Case II :



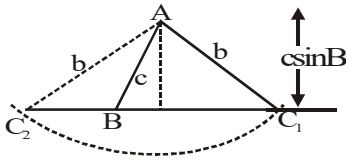
b = c sin B and B is an acute angle, there is only one triangle possible. and it is right-angled at C.

Case III :



c > b > c sin B, and B is an acute angle, then there are two triangles possible for two values of angle C.

Case IV :

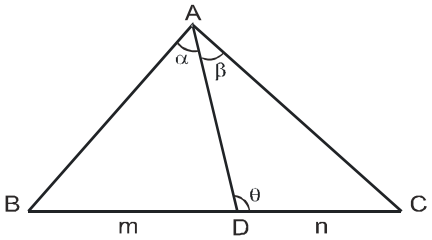


$b \geq c$, then there is only one triangle.

	Conditions	**No. of Δ exists**
(1)	$b < c\sin B$	0
(2)	$b = c\sin B$	1 ($\angle C = 90^\circ$)
(3)	$c > b > c\sin B$	2
(4)	$b \geq c$	1

4.2 m-n THEOREM

Let D be a point on the side BC of a Δ ABC such that



BD : DC = m : n and
$\angle ADC = \theta$, $\angle BAD = \alpha$ and $\angle DAC = \beta$.
Then :

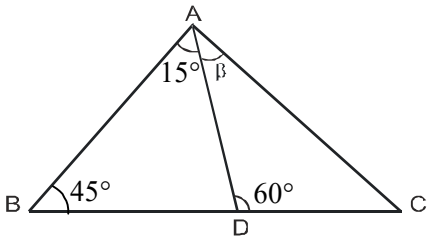
(i) $(m + n) \cot \theta = m \cot \alpha - n \cot \beta$

(ii) $(m + n) \cot \theta = n \cot B - m \cot C$

Questions

1. In ΔABC $\angle ABC = 45^\circ$, D is on BC such that $2BD = CD$ and $\angle DAB = 45^\circ$. Find $\angle ACB$.

Sol.



$2BD = CD$

$$\frac{BD}{CD} = \frac{1}{2}$$

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$$

$3\cot 60 = \cot 15^\circ - 2\cot \beta$

$$3 \times \frac{1}{\sqrt{3}} = 2 + \sqrt{3} - 2\cot B$$

$$\sqrt{3} = 2 + \sqrt{3} - 2\cot B$$

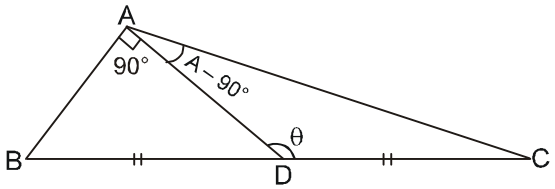
$\cot B = 1$

$B = 45^\circ$

$\angle A = 60^\circ$ & $\angle C = 75^\circ$

2. If median of ΔABC through A is perpendicular to AB then prove that $\frac{\tan A}{\tan B} = -2$.

Sol. From the figure, we see that $\theta = 90^\circ + B$ (as θ is external angle of ΔABD)



Now if we apply **m-n rule** in ΔABC, we get

$(1 + 1) \cot (90^\circ + B) = 1. \cot 90^\circ - 1.\cot (A - 90^\circ)$

$\Rightarrow \quad -2 \tan B = \cot (90^\circ - A)$

$\Rightarrow \quad -2 \tan B = \tan A$

$\Rightarrow \quad \tan A + 2 \tan B = 0$

3. In ΔABC AD is altitude $\angle BAC = 45^\circ$, $BD = 3$, $CD = 2$. Find area of ΔABC.

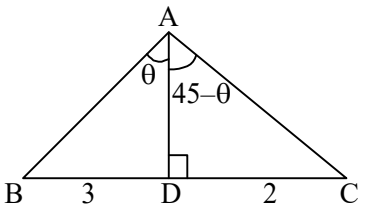
Sol. $\tan\theta = \frac{3}{AD} \Rightarrow AD = 3\cot\theta$

$$\text{Area } (\Delta) = \frac{1}{2} \times 5 \times 3\cot\theta$$

$$= \frac{15}{2}\cot\theta$$

$(3+2)\cot 90° = 3\cot\theta - 2\cot(45-\theta)$

$3\cot\theta = 2\cot(45-\theta)$



$$\frac{3}{\tan\theta} = \frac{2(1+\tan\theta)}{(1-\tan\theta)}$$

$\tan\theta = \frac{1}{2}, \tan\theta = -3$ (Reject $\theta < 45°$)

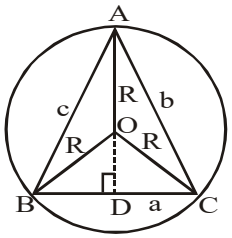
$\cot\theta = 2$

$\Delta = \frac{15}{2} \times 2 = 15$

5.1 CIRCUMCIRCLE AND CIRCUMRADIUS

The circle which passes through the angular points of a triangle is called its circumcircle. In a triangle the point of intersection of perpendicular bisector of the sides and is called the circumcentre. Its radius is always denoted by R. Circumcenter is equidistant from all three vertices.

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}.$$

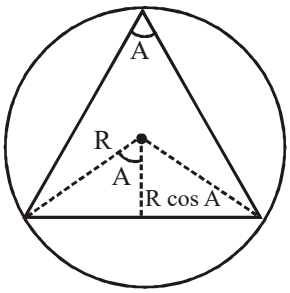


The circumcentre lies

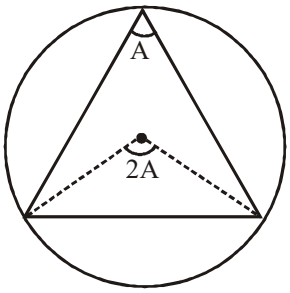
(1) inside an acute angled triangle (2) outside an obtuse angled triangle &

(3) mid point of the hypotenuse of right angled triangle.

Distance of circumcentre from sides

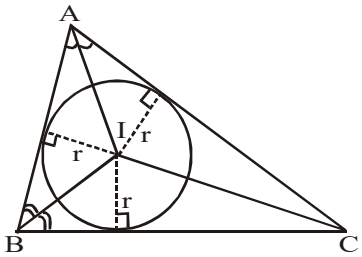


Angle subtended by sides at circumcentre

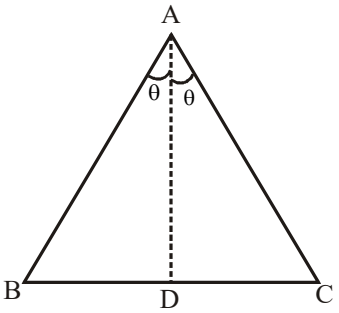


5.2 INCIRCLE AND INRADIUS

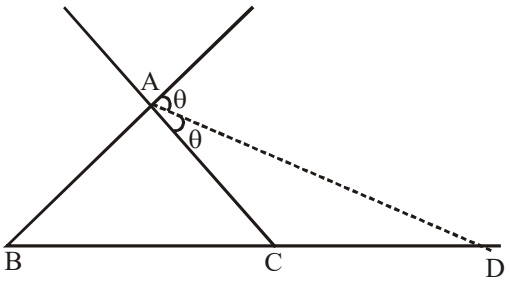
The circle which can be inscribed with in a triangle and touch each of the sides is called its inscribed circle or incircle. The centre of this circle is the point of intersection of the bisector of the angle of the triangle.
The radius of this circle is always denoted by r and is equal to the length of the perpendicular from its centre to any one of the sides of triangle.



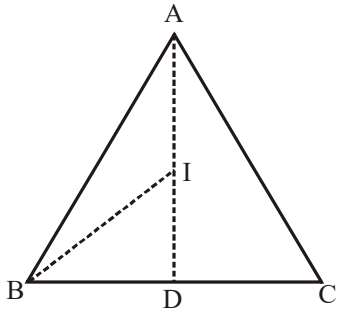
Incentre always lies inside the triangle.Incentre is equidistant from all the three sides. I is point of intersection of internal angle bisector of all the three angles.
Internal angle bisector divides the opposite in ratio of adjacent sides internally while external angle bisector does it externally.



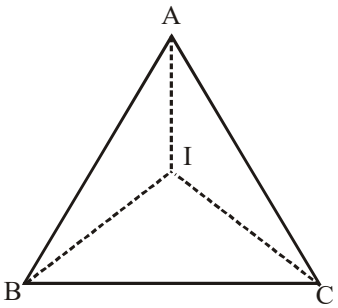
$$\frac{BD}{DC}=\frac{AB}{AC}=\frac{c}{b}$$



$$\frac{B'D}{D'C}=\frac{c}{b}$$



$$\frac{IA}{ID} = \frac{b+c}{a}$$



$$\Delta ABC = \Delta IBC + \Delta IAC + \Delta IAB$$

$$\Delta = \left(\frac{1}{2}a \times r\right) + \left(\frac{1}{2}b \times r\right) + \left(\frac{1}{2}c \times r\right)$$

$$\Delta = \frac{r}{2}(a+b+c)$$

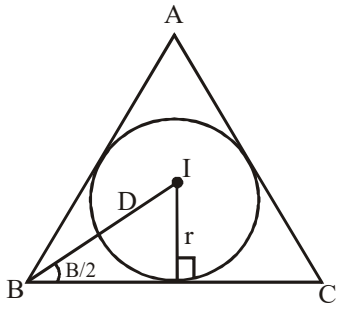
$$\Delta = \frac{r}{2}(2s)$$

(i) $r = \dfrac{\Delta}{s}$

(ii) $r = (s-a)\tan\left(\dfrac{A}{2}\right)$, $r = (s-b)\tan\left(\dfrac{B}{2}\right)$ and $r = (s-c)\tan\left(\dfrac{C}{2}\right)$

(iii) $r = 4R\sin\left(\dfrac{A}{2}\right).\sin\left(\dfrac{B}{2}\right).\sin\left(\dfrac{C}{2}\right)$

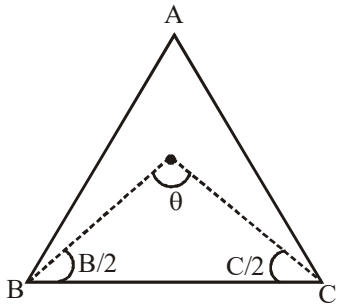
Distance of I from vertices



$$\sin\left(\frac{B}{2}\right) = \frac{r}{D}$$

$$IB = D = r\,\text{cosec}\,\frac{B}{2}; \quad IA = r\,\text{cosec}\,\frac{A}{2}; \quad IC = r\,\text{cosec}\,\frac{C}{2}$$

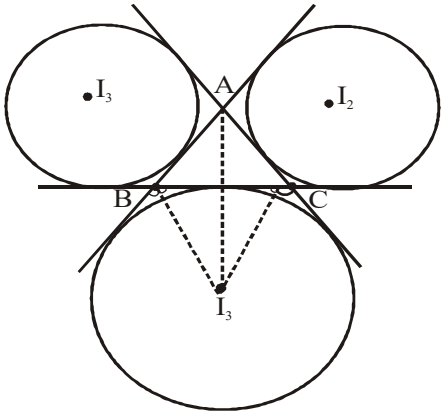
Angle subtended by side at incentre.



$$\theta = \pi - \frac{B}{2} - \frac{C}{2} \Rightarrow \theta = \pi - \frac{\pi - A}{2} = \frac{\pi}{2} + \frac{A}{2}$$

5.3 EXCIRCLES AND EXRADII -1

Point of intersection of two external angles and one internal angle bisectors is excentre and perpendicular distance of excentre from any side is called exradius. If r_1 is the radius of escribed circle opposite to $\angle A$ of ΔABC and so on, then -



I_1 = excentre w.r.t A.

I_2 = excentre w.r.t B.

I_3 = excentre w.r.t C.

r_1 = exradius w.r.t A.

r_2 = exradius w.r.t B.

r_3 = exradius w.r.t C.

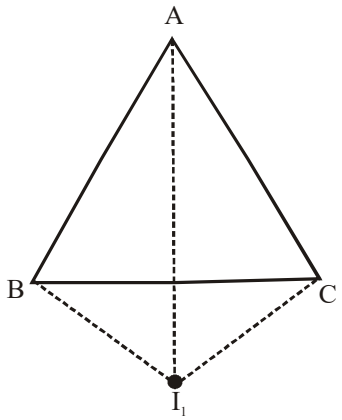
I_1 is at equal distance from all three sides (distance equal to r_1)

I_1 is point of intersection of internal angle bisector through A & external angle bisector through B & C

$$\Delta ABC + \Delta I_1BC = \Delta I_1CA + \Delta I_1AB$$

$$\Delta + \frac{1}{2}ar_1 = \frac{1}{2}br_1 + \frac{1}{2}cr_1$$

$$\Delta = \frac{1}{2}r_1(2s - 2a)$$



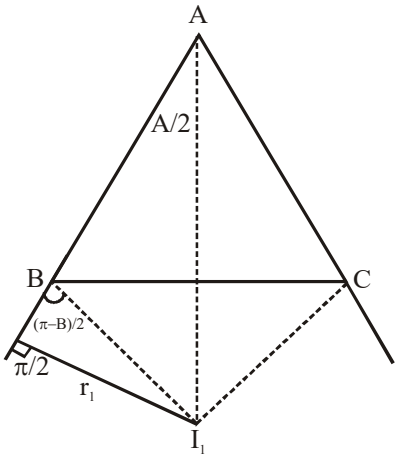
$$r_1 = \frac{\Delta}{s-a}; \qquad r_2 = \frac{\Delta}{s-b} \qquad r_3 = \frac{\Delta}{s-c}$$

$$r_1 = \frac{\Delta}{s-a} = s\tan\frac{A}{2} = 4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2};$$

$$r_2 = \frac{\Delta}{s-b} = s\tan\frac{B}{2} = 4R\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2};$$

$$r_3 = \frac{\Delta}{s-c} = s\tan\frac{C}{2} = 4R\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$$

Distance of excenters from vertices



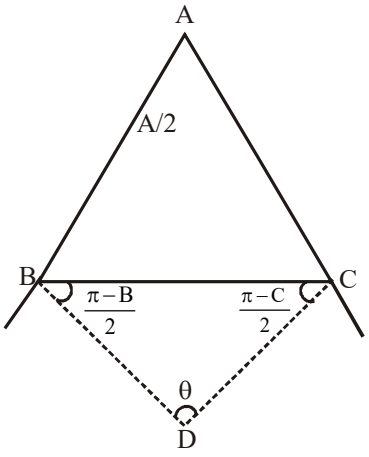
$$\sin\frac{A}{2}=\frac{r_1}{I_1A} \qquad \cos\frac{B}{2}=\frac{r_1}{I_1B}$$

$$I_1A=r_1\cos ec\frac{A}{2} \qquad I_1B=r_1\sec\frac{B}{2} \qquad I_1C=r_1\sec\frac{C}{2}$$

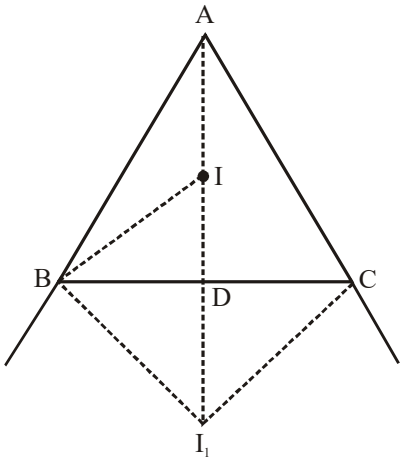
$$I_2B=r_2\operatorname{cosec}\frac{B}{2} \qquad I_2A=r_2\sec\frac{A}{2} \qquad I_2C=r_2\sec\frac{C}{2}$$

$$I_3C=r_3\operatorname{cosec}\frac{C}{2} \qquad I_3A=r_3\sec\frac{A}{2} \qquad I_3B=r_3\sec\frac{B}{2}$$

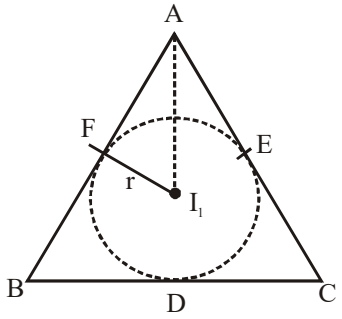
Angle subtended by sides at excenters

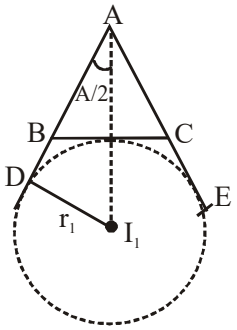


$$\theta=\pi-\left(\frac{\pi}{2}-\frac{B}{2}+\frac{\pi}{2}-\frac{C}{2}\right)=\frac{B+C}{2}=\frac{\pi}{2}-\frac{A}{2}$$



$$\frac{I_1A}{I_1D} = \frac{IA}{ID} = \frac{b+c}{a}$$





$AF = AE = s - a$

$BF = BD = s - b$

$CE = CD = s - c$

$AD = AE = s$

Questions

1. Prove that $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$.

Sol. $\frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta}$

$= \frac{3s-a-b-c}{\Delta}$

$= \frac{s}{\Delta}$

$= \frac{1}{r}$

2. Prove that $r_1r_2 + r_2r_3 + r_3r_1 = s^2$.

Sol. $s\tan\frac{A}{2} \times s\tan\frac{B}{2} + s\tan\frac{B}{2} \times s\tan\frac{C}{2} + s\tan\frac{C}{2} \times s\tan\frac{A}{2}$

$$s^2\left(\tan\frac{A}{2}.\tan\frac{B}{2}+\tan\frac{B}{2}.\tan\frac{C}{2}+\tan\frac{C}{2}.\tan\frac{A}{2}\right)$$

$s^2 \times 1 = s^2$

3. Prove that $r.\, r_1.\, r_2.\, r_3 = \Delta^2$.

Sol. $$\frac{\Delta}{s}\times\left(\frac{\Delta}{s-a}\right)\times\left(\frac{\Delta}{s-b}\right)\times\left(\frac{\Delta}{s-c}\right)$$

$$=\frac{\Delta^4}{s(s-a)(s-b)(s-c)}=\Delta^2$$

4. Prove that $r_1+r_2+r_3-r=4R$.

Sol. $\because$ L.H.S $= r_1 + r_2 + r_3 - r$

$$= \frac{\Delta}{s-a} + \frac{\Delta}{s-b} + \frac{\Delta}{s-c} - \frac{\Delta}{s}$$

$$= \Delta\left(\frac{1}{s-a}+\frac{1}{s-b}\right) + \Delta\left(\frac{1}{s-c}-\frac{1}{s}\right) = \Delta\left[\left(\frac{s-b+s-a}{(s-a)(s-b)}\right)+\left(\frac{s-s+c}{s(s-c)}\right)\right]$$

$$= \Delta\left[\frac{c}{(s-a)(s-b)}+\frac{c}{s(s-c)}\right] = c\Delta\left[\frac{s(s-c)+(s-a)(s-b)}{s(s-a)(s-b)(s-c)}\right]$$

$$= c\Delta\left[\frac{2s^2-s(a+b+c)+ab}{\Delta^2}\right] = \frac{abc}{\Delta} \quad \because \quad a+b+c = 2s$$

$$\because \quad R = \frac{abc}{4\Delta}$$

$= 4R = 2a \operatorname{cosec} A$ $\quad \because \quad \frac{a}{\sin A} = 2R = a \operatorname{cosec} A$

$= R.H.S.$

Hence L.H.S. $=$ R.H.S. **proved**

5. In ΔABC $a = 13, b = 30, c = 37$

Find (i) Δ, (ii) r, (iii) R, (iv) r_1, r_2, r_3

Sol. $$s=\frac{a+b+c}{2}=40$$

(i) $$\Delta=\sqrt{s(s-a)(s-b)(s-c)}=\sqrt{40\times27\times10\times3}=180$$

(ii) $$r=\frac{\Delta}{s}=\frac{180}{40}=\frac{9}{2}$$

(iii) $$R=\frac{abc}{4\Delta}=\frac{30.13.37}{4\times180}=\frac{481}{24}$$

(iv) $$r_1=\frac{\Delta}{s-a}=\frac{180}{40-13}=\frac{180}{27}=\frac{20}{3}$$

$$r_2=\frac{\Delta}{s-b}=\frac{180}{40-30}=\frac{180}{10}=18$$

$$r_3 = \frac{\Delta}{s-c} = \frac{180}{40-37} = \frac{180}{3} = 60$$

6. Prove that $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.

Sol. $\cos A + \cos B + \cos C = 2\cos\left(\frac{A+B}{2}\right).\cos\left(\frac{A-B}{2}\right) + \cos C$

$$= 2\sin\frac{C}{2}.\cos\left(\frac{A-B}{2}\right) + 1 - 2\sin^2\frac{C}{2} = 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \sin\left(\frac{C}{2}\right)\right]$$

$$= 1 + 2\sin\frac{C}{2}\left[\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right)\right] \qquad \left\{\because \frac{C}{2} = 90° - \left(\frac{A+B}{2}\right)\right\}$$

$$= 1 + 2\sin\frac{C}{2}.2\sin\frac{A}{2}.\sin\frac{B}{2} = 1 + 4\sin\frac{A}{2}.\sin\frac{B}{2}.\sin\frac{C}{2}$$

$$= 1 + \frac{r}{R} \qquad \{\text{as, } r = 4R\sin A/2 \,.\, \sin B/2 \,.\, \sin C/2\}$$

$\Rightarrow$ $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$. Hence proved.

7. For ΔABC if $r_1 = 8$, $r_2 = 12$, $r_3 = 24$ find.
(i) r (ii) Δ (iii) s (iv) R (v) a,b,c

Sol. $\because$ $\Delta = 96$ sq. unit

$r_1 = 8$, $r_2 = 12$ and $r_3 = 24$

$\because$ $r_1 = \frac{\Delta}{s-a}$ $\Rightarrow$ $s - a = 12$ .....(i)

$\because$ $r_2 = \frac{\Delta}{s-b}$ $\Rightarrow$ $s - b = 8$ .....(ii)

$\because$ $r_3 = \frac{\Delta}{s-c}$ $\Rightarrow$ $s - c = 4$ .....(iii)

$\therefore$ adding equations (i), (ii) & (iii), we get

$3s - (a + b + c) = 24$

$s = 24$

$\therefore$ perimeter of $\Delta ABC = 2s = 48$ unit. Ans.

8. Prove that $a = \frac{r_1(r_2 + r_3)}{\sqrt{\sum r_1 r_2}}$.

Sol. RHS $\frac{r_1(r_2 + r_3)}{\sqrt{r_1r_2 + r_2r_3 + r_3r_1}}$

$$= \frac{s\tan\frac{A}{2}\left(s\tan\frac{B}{2} + s\tan\frac{C}{2}\right)}{\sqrt{s^2}}$$

$$= s\tan\frac{A}{2}\left(\tan\frac{B}{2} + \tan\frac{C}{2}\right)$$

$$= s \times \frac{\Delta}{(s-a)}\left(\frac{\Delta}{s(s-b)} + \frac{\Delta}{s(s-c)}\right)$$

$$= s\left(\frac{\Delta^2}{s(s-a)(s-b)} + \frac{\Delta^2}{s(s-a)(s-c)}\right)$$

$= (2s - c - b) = a$

9. If $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right) = 2$ prove that triangle is right angled.

Sol. $\left[1-\frac{(s-b)}{(s-a)}\right]\left[1-\frac{(s-c)}{(s-a)}\right] = 2$

$(r_2 - r_1)(r_3 - r_1) = 2r_2r_3$

$r_1^2 = r_1r_2 + r_2r_3 + r_3r_1$

$r_1^2 = s^2 \Rightarrow r_1 = s$

$s\tan\frac{A}{2} = s$

$\frac{A}{2} = \frac{\pi}{4} \Rightarrow A = \frac{\pi}{2}$

10. Prove that $a\cos A + b\cos B + c\cos C = 4R\sin A\sin B\sin C$.

Sol. $a\cos A + b\cos B + c\cos C$

$= 2R\sin A\cos A + 2R\sin B\cos B$

$+ 2R\sin C\cos C$

$[\because a = 2R\sin A, b = 2R\sin B, c = 2R\sin C]$

$= R(\sin 2A + \sin 2B + \sin 2C)$

$= R[2\sin(A + B)\cos(A - B) + 2\sin C\cos C]$

$[\because A + B + C = \pi]$

$= R[2\sin C\cos(A - B) + 2\sin C\cos C]$

$= 2R\sin C[\cos(A - B) + \cos C]$

$= 2R\sin C[\cos(A - B) + \cos(\pi - (A + B))]$

$= 2R\sin C[\cos(A - B) - \cos(A + B)]$

$= 2R\sin C[2\sin A\sin B]$

$= 4R\sin A\sin B\sin C$

11. Prove that $a\cos B\cos C + b\cos C\cos A + c\cos A\cos B = \frac{\Delta}{R}$.

Sol. $a\cos B\cos C + \cos A(b\cos C + c\cos B) = \frac{ab\sin C}{2\times b} \times 2\sin B$

$a\cos B\cos C + a\cos A = a\sin C.\sin B$

$\cos B\cos C - \sin C.\sin B = -\cos A$

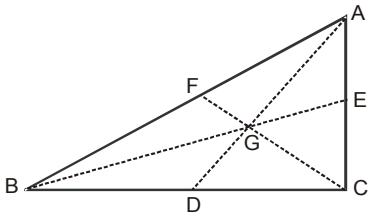
$\cos(B + C) = -\cos A$

$-\cos A = -\cos A$

6.1 CENTROID AND MEDIAN

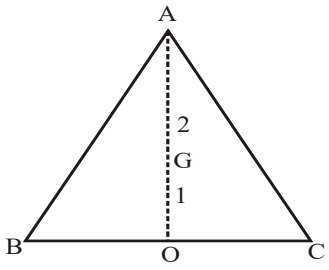
The line joining any vertex of a triangle to the mid point of the opposite side of the triangle is called the median of the triangle. The three medians of a triangle are concurrent and the point of concurrency of the medians of any triangle is called the centroid of the triangle.

In the given triangle ABC , D, E and F respectively are mid points of BC, AC, AB and the lines AD, BE, CF are called the medians and the intersection point G is called the centroid.

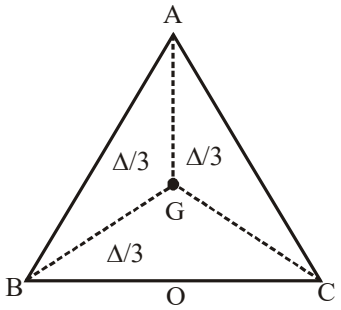


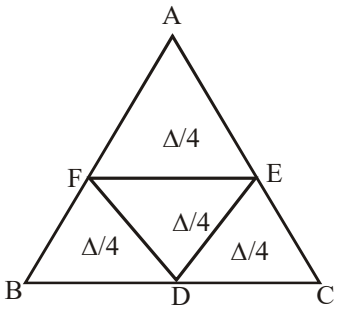
Centroid always lies inside the triangle. Median divides a triangle in two equal areas.

Centroid divides the median in 2:1 ratio internally

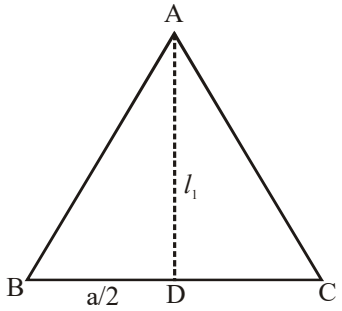


$$\frac{GA}{GD} = \frac{2}{1}$$





6.2 LENGTH OF MEDIAN



$$\ell_1 = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

Proof:– In ΔABC

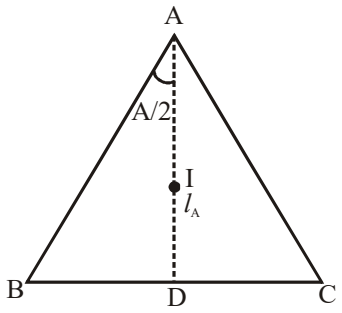
$$\cos B = \frac{c^2 + a^2 - b^2}{2ac}$$

In ΔABD

$$\cos B = \frac{c^2 + \left(\frac{a}{2}\right)^2 - \ell_1^2}{2c\left(\frac{a}{2}\right)}$$

$$\frac{c^2 + a^2 - b^2}{2ac} = \frac{c^2 + \frac{a^2}{4} - \ell_1^2}{ac}$$

6.3 LENGTH OF ANGLE BISECTOR



$$\ell_A = \frac{2bc}{b+c}\cos\frac{A}{2}$$

Proof:–

Area of $\Delta ABC = \Delta ABD + \Delta ACD$

$$\frac{1}{2}bc\sin A = \frac{1}{2}\ell_A(AB)\sin\frac{A}{2} + \frac{1}{2}\ell_A(AC)\sin\frac{A}{2}$$

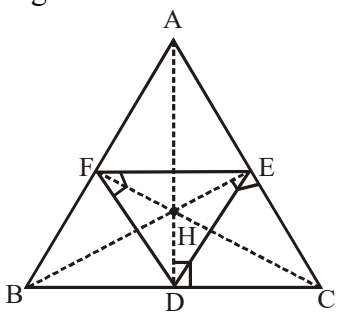
$$bc\sin A = \ell_A(b+c)\sin A/2$$

$$2bc\sin\frac{A}{2}\cos\frac{A}{2} = \ell_A(b+c)\sin\frac{A}{2}$$

$$\ell_A = \frac{2bc}{b+c}\cos\frac{A}{2}$$

7. ORTHOCENTER AND PEDAL TRIANGLE

Orthocentre is point of intersection(POI) of altitudes. Triangle formed by joining feet of altitudes is called pedal triangle.



AD, BE, CF $\rightarrow$ altitudes; D, E, F feet of altitude

$\Delta DEF =$ pedal triangle of ΔABC

Distance of orthocentre from sides & vertices

In ΔABD

$$\cos B = \frac{BD}{c}$$

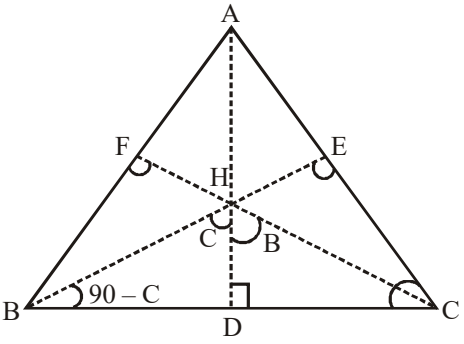
$$BD = c\cos B$$

In ΔHBD

$$\sin C = \frac{BD}{HB}$$

$$HB = \frac{BD}{\sin C} = \frac{c\cos B}{\sin C}$$

$HB = 2R\cos B$; $HA = 2R\cos A$; $HC = 2R\cos C$



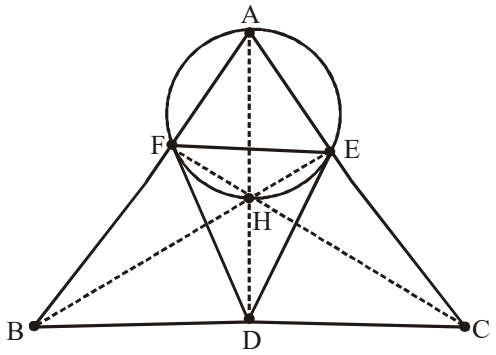
$$\cos C = \frac{HD}{HB}$$

$$HD = HB\cos C$$

$HD = 2R\cos B\cos C$; $HE = 2R\cos C\cos A$; $HF = 2R\cos A\cos B$

Angle subtended by BC at H = B + C $= \pi - A$

Sides and angles of pedal triangle



AFHE is cylic quadrilateral. The circle circumscribing it is also the circumcircle of ΔAEF, for this circle HA is diameter.

In ΔAEF by sine rule.

$$\frac{EF}{\sin A} = 2R_{AEF} = AH$$

$$EF = 2R\cos A\sin A$$
$$= R\sin 2A$$

$EF = a' = R\sin 2A = a\cos A$; $FD = b' = R\sin 2B = a\cos B$; $ED = c' = R\sin 2C = a\cos C$

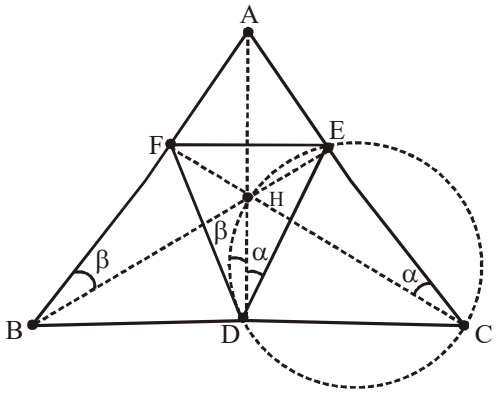
for circle HE is chord

$\angle HDE = \angle HCE = 90^\circ - A$

$\angle HDF = \angle HBE = 90^\circ - A$

$\angle FDE = \alpha + \beta = \pi - 2A$

$A^1 = \pi - 2A$; $B^1 = \pi - 2B$; $C^1 = \pi - 2C$



HD, HE & HF are internal angle bisectors of ΔDEF so orthocentre of ΔABC is incentre of pedal triangle DEF.

Circumradius of pedal triangle DEF is R/2 where R is circumradius of ΔABC

Proof:– $2R_P = \frac{a'}{\sin A'}$

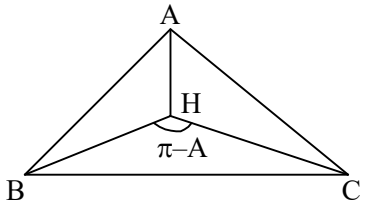
$$2R_P = \frac{R \sin 2A}{\sin(\pi - 2A)}$$

$$2R_P = R$$

Questions

1. Prove that circumradius of ΔABC, ΔHBC, ΔHAC, ΔHAB are equal.

Sol.



In ΔHBC

$$2R_{HBC} = \frac{a}{\sin(\pi - A)}$$

$2R_{HBC} = 2R$

$R_{HBC} = R$

2. Prove that $\Delta_p = 2\Delta \cos A \cos B \cos C$.

Sol. $\Delta_p = \frac{1}{2} a'b' \sin(\pi - 2c)$

$$= \frac{1}{2} \times ab \cos A \cos B \sin 2C$$

$$= \frac{1}{2} \times ab \times 2 \sin C \cos C \cos A \cos B$$

$= 2\Delta \cos A \cos B \cos C$

3. Prove that $r_p = 2R \cos A \cos B \cos C$.

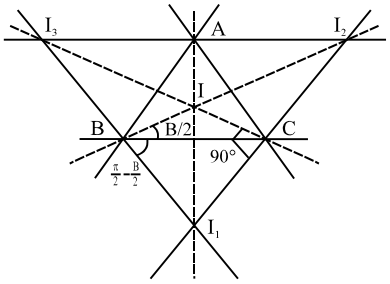
Sol. $r_P = 4R_P \sin\left(\frac{A'}{2}\right) \sin\left(\frac{B'}{2}\right) \sin\left(\frac{C'}{2}\right)$

$r_p = 4R_p \cos A \cos B \cos C$

Put $R_p = \frac{R}{2}$

$r_p = 2R \cos A \cos B \cos C$

8.1 EXCENTRAL TRIANGLE



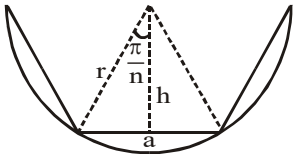
The triangle formed by joining the three excentres I_1, I_2 and I_3 of Δ ABC is called the excentral or excentric triangle.

I_1A is internal angle bisector & I_2I_3 is external angle bisector for $\angle A$, so IA is perpendicular to I_2I_3. So for $\Delta I_1I_2I_3$, A, B, C are feet of altitudes. ΔABC is pedal triangle of $\Delta I_1I_2I_3$.

Pedal triangle	**Excentral triangle**
a, b, c	I_2I_3, I_3I_1, I_1I_2
a', b', c'	a, b, c
A, B, C	$\angle I_1$, $\angle I_2$, $\angle I_3$
A', B', C'	A, B, C
$A' = \pi - 2A$	$A = \pi - 2I_1$
	$I_1 = \frac{\pi}{2} - \frac{A}{2};\ I_2 = \frac{\pi}{2} - \frac{B}{2};\ I_3 = \frac{\pi}{2} - \frac{C}{2}$
$a' = a\cos A$	$a = I_2I_3\cos(I_1)$
	$a = I_2I_3\cos\left(\frac{\pi}{2} - \frac{A}{2}\right)$
	$I_2I_3 = a\,\text{cosec}\frac{A}{2};\ I_3I_1 = b\,\text{cosec}\frac{B}{2};\ I_1I_2 = c\,\text{cosec}\frac{C}{2}$

8.2 REGULAR POLYGON

A regular polygon has all its sides equal. It may be inscribed or circumscribed.

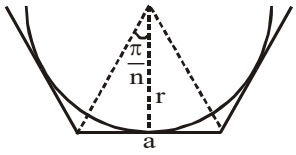


(a) Inscribed in circle of radius r :

(i) $a = 2h\ \tan\frac{\pi}{n} = 2r\sin\frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr\sin\frac{\pi}{n}$ and $A = \frac{1}{2}nr^2\sin\frac{2\pi}{n}$

(b) Circumscribed about a circle of radius r :



(i) $a = 2r \tan \frac{\pi}{n}$

(ii) Perimeter (P) and area (A) of a regular polygon of n sides circumscribed about a given circle of radius r is given by $P = 2nr \tan \frac{\pi}{n}$ and $A = nr^2 \tan \frac{\pi}{n}$

8.3 IMPORTANT POINTS :

1. In right angle triangle $\cos^2 A + \cos^2 B + \cos^2 C = 1$

2. In equilateral triangle

(i) $R = 2r$ (ii) $r_1 = r_2 = r_3 = \frac{3R}{2}$ (iii) $r : R : r_1 = 1 : 2 : 3$ (iv) area $= \frac{\sqrt{3}a^2}{4}$ (v) $R = \frac{a}{\sqrt{3}}$

3. Area of a cyclic quadrilateral $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$

where a, b, c, d are lengths of the sides of quadrilateral and $s = \frac{a+b+c+d}{2}$.

4. Distance Between Special Points :

(i) Distance between circumcentre and orthocentre $OH^2 = R^2 (1 - 8 \cos A \cos B \cos C)$

(ii) Distance between circumcentre and incentre $OI^2 = R^2 (1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}) = R^2 - 2Rr$

(iii) Distance between circumcentre and centroid $OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$